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Glare: A free and open-source software for generation and assessment of digital speckle pattern

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ABSTRACT

Generating digital speckle image and its corresponding deformed image is the basis of digital image correlation research. At present, however, it still lacks a powerful, easy-to-use, and user-friendly professional software concerning generation and assessment of digital speckle pattern. Researchers have to reimplement the generation algorithms in literature by themselves, which is time-consuming and error-prone. This paper reports a free and open-source software, *Glare*, for generation and assessment of digital speckle pattern. *Glare* has functions including generating speckle patterns, rendering deformed images, assessing pattern quality, and presenting pattern recommendations: *Glare* can generate ellipse, polygon, and Gaussian speckle patterns; can render deformed images with underlying deformation fields of translation, stretch/compression, rotation, sinusoidal deformation, Gaussian deformation, and Portevin-Le Chatelier band deformation; can calculate key pattern quality assessment parameters such as speckle coverage, speckle size, systematic error, and random error; can produce optimized speckle pattern in form of vector image. The software realizes real-time deformed image rendering with the aid of fast initial value estimation algorithm for backward mapping and pattern pre-rendering technique, and improves the computational efficiency of sum of square of subset intensity gradients by integral image method. In general, the software can be used not only for scientific research and engineering applications in digital image correlation community but also for education of experimental mechanics, and therefore has broad prospects.

1. Introduction

Digital image correlation (DIC) is a non-contacting and non-interferometric full-field optical metrology. Since its invention in the early 1980s [1,2], DIC has been widely used in various fields such as material testing, aerospace industry, biomechanics, and civil engineering, because of its advantages such as simple equipment, low environmental requirement, and high accuracy. In recent years, with the presentation of inverse strategy [3], GPU-based acceleration [4], and convolutional neural network StrainNet [5], the computational efficiency of DIC has been greatly improved and real-time measurement was realized [6], further broadening its application scope.

The principle of DIC is to register the correspondence of the same physical point from different moments, different views, and different sensors through the texture of the specimen surface [7]. Hence, the texture has a profound impact on the metrological performance of DIC. In DIC community, the texture has a specific terminology, termed *speckle pattern*.

Generating digital speckle image and its corresponding deformed image is the basis of digital image correlation research. Scholars have proposed many types of speckle patterns, including digital speckle pattern [8], Boolean model speckle pattern [9], checkerboard speckle pattern [10], Perlin's coherent noise based speckle pattern [11], etc. Among them, digital speckle pattern is extremely popular for its simpleness, controllable randomness, and convenient fabrication. Recently, improving metrological performance of DIC by speckle pattern optimization becomes a research hotspot. The foundation of speckle optimization is comprehensive quality assessment of speckle pattern, and various speckle pattern quality assessment parameters have been presented, such as speckle size [12], speckle coverage [13], mean image gradient [14], and root mean square error [15]. Based on these assessment parameters, a series of studies on optimization of digital speckle patterns have been conducted: Lavatelli et al[16]. developed a closed-loop pattern optimization engine based on 3-D experiment simulation; Chen et al[17]. optimized generation parameters of digital speckle pattern through numerical simulations; Su et al[18]. established theoretical models for speckle uniqueness, spatial resolution, systematic error, random error, and total error, and presented a strategy to select the optimal pattern generation parameters.

Although speckle image generation, deformed image rendering, speckle pattern assessment, and optimal speckle pattern generation have been thoroughly studied, it still lacks a powerful, easy-to-use, and user-

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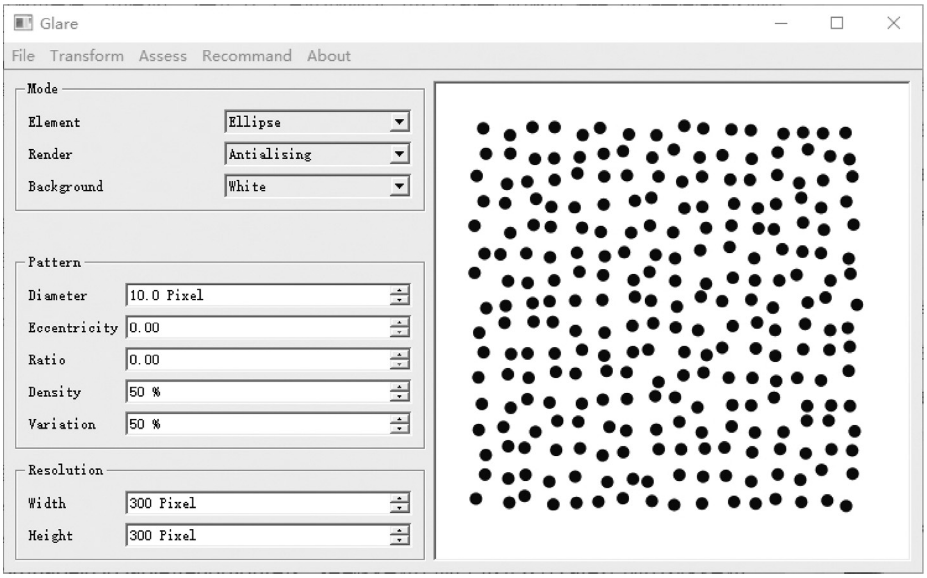

Fig. 1. Graphical user interface of *Glare*.

friendly professional software concerning generation and assessment of digital speckle pattern. Speckle Generator [19], a software developed by Correlated Solutions, can produce speckle patterns composed of randomly positioned disks; although it is pioneering and illuminating, its functionality is rather limited: it cannot produce bitmap image, cannot render deformed image, cannot assess pattern quality, and cannot recommend generation parameters; moreover, as a proprietary software, it is closed source and hence cannot be modified. BSpeckleRender [9], a powerful MATLAB software based on Boolean model, can render speckle images that are quite similar to realistic ones; however, its calculation burden is relatively heavy and the displacement field is described by the coordinates in the deformed image rather than the reference image. Hence, researchers generally have to reimplement the generation algorithms in literature by themselves, which is time-consuming and error-prone.

To tackle these issues, we developed a free and open-source software, *Glare*, for generation and assessment of digital speckle pattern. The primary aim of developing *Glare* is to serve academic research, engineering applications, and teaching of DIC. Being a software with graphical user interface (see Fig. 1), *Glare* has functions including generating speckle patterns, rendering deformed images, assessing pattern quality, and presenting pattern recommendations. The software is written in C++ together with Qt [20], OpenCV [21], Armadillo [22], and QCustomPlot [23]. The homepage of the software is https://gitee.com/yongsu1989/glare, under GPL 3.0 license.

2. Speckle pattern generation

2.1. Digital speckle pattern and digital speckle image

Digital speckle pattern [8,17], which is composed of randomly positioned speckles, has generation parameters including speckle profile $\varphi(x, y)$, speckle density ρ, and speckle variation δ; its mathematical form is as follows:

$$f(x, y) = \sum_{i=1}^{N} \sum_{j=1}^{N} \varphi(x - x_{i,j}, y - y_{i,j}), \tag{1}$$

where $\varphi(x, y)$ denotes the profile of an individual speckle, vector $(x_{i,j}, y_{i,j})$ denotes the centroid of the speckle at the ith row and jth column, and the total number of speckles is N^2.

The generation process of digital speckle pattern is shown in Fig. 2: first, according to given diameter D and density ρ, a grid with a step of $L = D/\rho$ is generated; then, speckles are positioned on the grid points, as

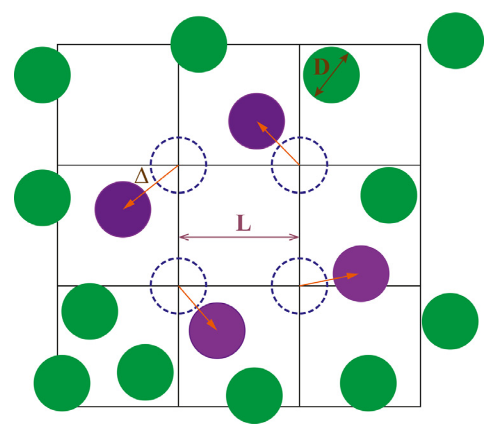

Fig. 2. The generation process of digital speckle pattern.

illustrated by the dashed lines in Fig. 2; finally, each speckle is imposed a random displacement $\Delta = (\Delta_x, \Delta_y)$, where random variables Δ_x and Δ_y satisfy a uniform distribution over interval $[-\delta L/2, \delta L/2]$, recalling that δ is the speckle variation. The imposed displacements are totally random; hence, even though the generation parameters are the same, each time a different speckle pattern will be generated.

Digital speckle image is obtained by sampling and quantizing digital speckle pattern. If the quantization range is $[g_b, g_f]$ and the maximum and the minimum of digital speckle pattern $f(x, y)$ are f_{max} and f_{min} respectively, the grayscale of digital speckle image is

$$g[m, n] = \mathrm{Round}\left[g_b + (g_f - g_b)\frac{f(m, n) - f_{min}}{f_{max} - f_{min}}\right]. \tag{2}$$

The bit depth of generated image is 8 bits; black background with white speckles corresponds to $g_b = 0$ and $g_f = 255$, while white background with black speckles corresponds to $g_b = 255$ and $g_f = 0$.

2.2. Ellipse, polygon, and Gaussian speckle images

Glare can generate ellipse, polygon, and Gaussian speckle images. Here, ellipse and polygon speckle images are rendered by Qt paint engine [20], while Gaussian speckle images are generated by evaluating the grayscale of each pixel through mathematical expression.

(1) For ellipse speckle images (Fig. 3(a1)-3(a4), the speckle radius r and the eccentricity e can be adjusted. In such a case, the semi-major

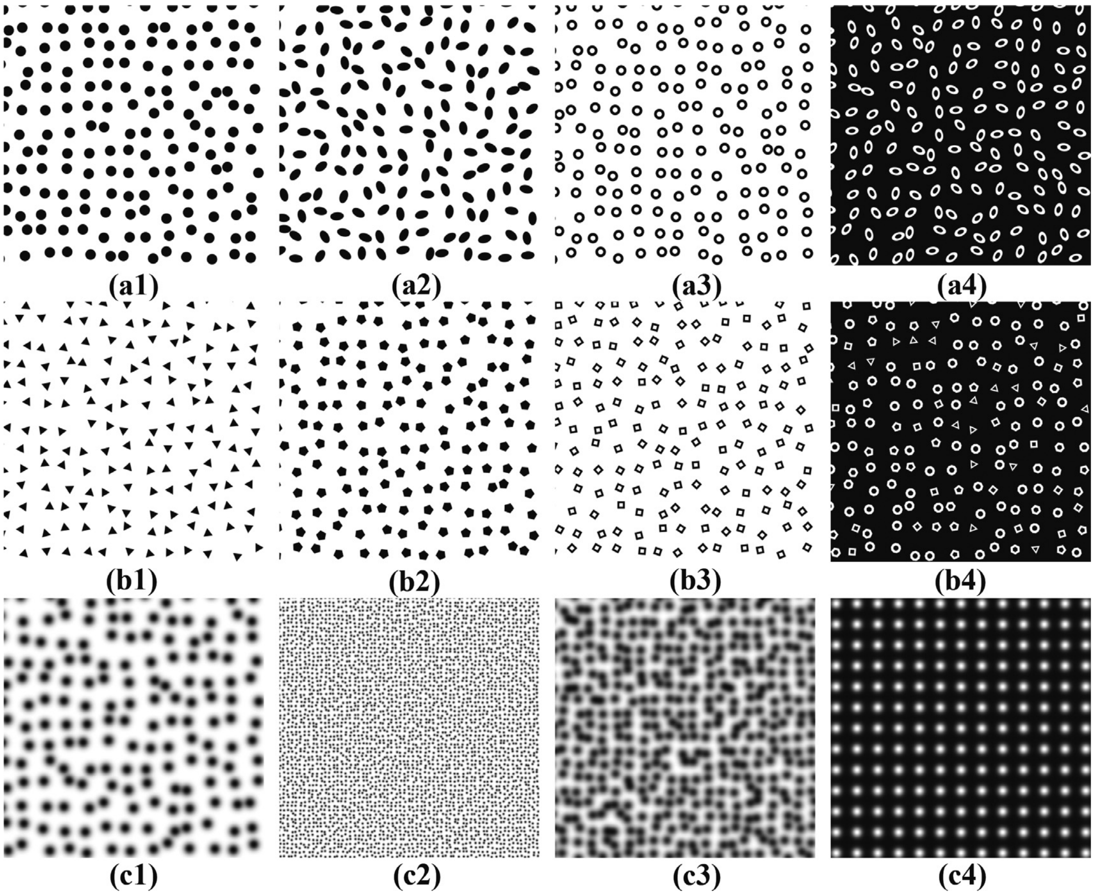

Fig. 3. Speckle patterns that *Glare* can generate: the first, second, and third rows are ellipse speckle patterns, polygon speckle patterns, and Gaussian speckle patterns, respectively.

axis a and the semi-minor axis b are

$$a = \frac{r}{\sqrt[4]{1-e^2}}, \qquad b = r\sqrt[4]{1-e^2}. \tag{3}$$

Thus, the area of the ellipse is $S = \pi ab = \pi r^2$, which is identical to corresponding circle. It can also be verified that the eccentricity of the ellipse is

$$\frac{\sqrt{a^2-b^2}}{a} = e. \tag{4}$$

To eliminate anisotropy, each ellipse speckle is randomly rotated (Fig. 3(a2)). Besides, *Glare* is able to generate hollow ellipse (Fig. 3(a3)-3(a4)): given ratio α, an elliptical hole, whose semi-major and semi-minor axes are αa and αb respectively, is generated inside each ellipse. Moreover, *Glare* can also generate image with black background and white speckles, and settings such as eccentricity, hollow ratio, and black background can be imposed simultaneously (Fig. 3(a4)).

(2) For polygon speckle images (Fig. 3(b1)-3(b4)), the polygon edges n and the circumradius r can be set. If $n < 3$, the polygons edges will be assigned randomly. Besides, settings such as hollow ratio (Fig. 3(b3)) and black background (Fig. 3(b4)) can be set.

(3) For Gaussian speckle images (Fig. 3(c1)-(c4)), the speckle profile satisfies [24]

$$\varphi(x,y) = \exp\left[-\frac{x^2+y^2}{r^2}\right], \tag{5}$$

where r is the radius of Gaussian speckle.

Compared with ellipse and polygon speckle patterns, Gaussian speckle pattern has relatively smooth grayscale distribution; meanwhile, if the point spread function of the optical system can be approximated by a Gaussian function, the image of a point light source can be approximated by a Gaussian speckle; therefore, Gaussian speckle pattern is extensively utilized in DIC research.

3. Deformed image rendering

3.1. Deformation mode

In order to analyze the performance of DIC algorithms, it is generally necessary to generate deformed image with given displacement field. To meet this need, *Glare* provides functions of rendering deformed image with deformation modes of translation, stretch/compression, rotation, sinusoidal deformation, Gaussian deformation, and Portevin-Le Chatelier (PLC) band deformation [25]. In future, more deformation modes will be added according to user feedback.

The principle of deformed image rendering is as follows: for an integer position (x', y') in the deformed image, find its corresponding position (x, y) in the reference image using the underlying displacement field (u, v), and then assign the grayscale via brightness constancy assumption. The key of deformed image rendering is to register the correspondence. The displacement field (u, v) is generally a function of reference image coordinate (x, y), and therefore the correspondence can be expressed as

$$x' = x + u(x,y), \qquad y' = y + v(x,y). \tag{6}$$

To simplify the discussion, *Glare* utilizes Gaussian speckle pattern and assumes the displacement is along the x axis exclusively (namely $v = 0$) except for rotation deformation.

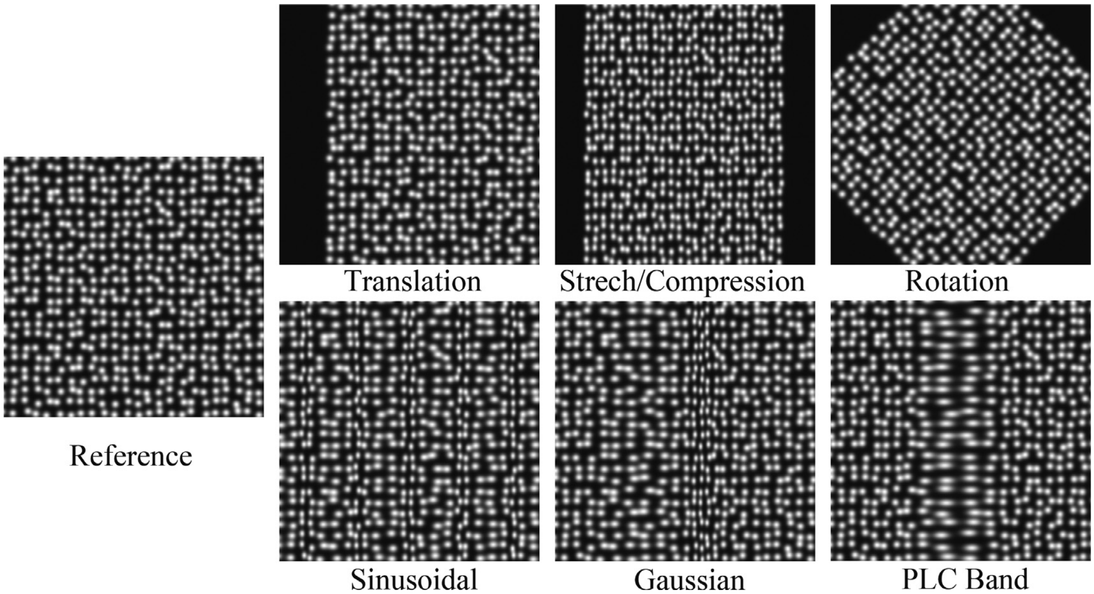

Fig. 4. Reference image and deformed images corresponding to different displacement fields including translation, stretch/compression, rotation, sinusoidal deformation, Gaussian deformation, and Portevin-Le Chatelier (PLC) band deformation.

(1) For translation displacement field (Fig. 4, right, row 1, column 1), the displacement u satisfies

$$u = u_0, \tag{7}$$

where u_0 is the translation.

(2) For stretch/compression deformation (Fig. 4, right, row 1, column 2), the displacement u satisfies

$$u = a(x - x_0), \tag{8}$$

where a characterizes the degree of stretch/compression, $a > 0$ indicates stretch and $a < 0$ indicates compression; x_0 denotes the center of stretching or compression.

(3) For rotation (Fig. 4, right, row 1, column 3), the reference position (x, y) and the deformed position (x', y') satisfies

$$\begin{bmatrix} x' - x_0 \\ y' - y_0 \end{bmatrix} = a \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}, \tag{9}$$

where θ denotes the rotation angle; (x_0, y_0) denotes the rotation center; a denotes the scale factor.

(4) For sinusoidal deformation (Fig. 4, right, row 2, column 1), the displacement u satisfies

$$u = a \sin\left(\frac{2\pi x}{T} + b\right), \tag{10}$$

where a denotes the amplitude of sinusoidal displacement; T denotes the period; b denotes the phase.

(5) For Gaussian deformation (Fig. 4, right, row 2, column 2), the displacement u satisfies

$$u = a \exp\left[-\frac{(x - x_0)^2}{c_g^2}\right], \tag{11}$$

where a denotes the maximum of Gaussian displacement field; x_0 denotes the center of Gaussian deformation; c_g characterizes the width the Gaussian deformation.

(6) For PLC band deformation (Fig. 4, right, row 2, column 3), the strain e_{xx} satisfies

$$e_{xx} = a \exp\left[-\frac{(x - x_0)^2}{c_{plc}^2}\right], \tag{12}$$

The difference between PLC band deformation and Gaussian deformation is that, the strain of PLC band deformation is Gaussian, while the displacement field of Gaussian deformation is Gaussian. Integrating Eq. (12) and introducing boundary condition $u = 0$ at $x = 0$ yield

$$u = \frac{1}{2} ac\sqrt{\pi}\left[erf\left(\frac{x_0}{c_{plc}}\right) + erf\left(\frac{x - x_0}{c_{plc}}\right)\right], \tag{13}$$

where erf denotes error function.

3.2. Backward mapping

Backward mapping means finding the corresponding reference image coordinate for a point in the deformed image. For linear deformations such as translation, stretch/compression, and rotation, it can be handled by solving linear equations. Nevertheless, for complex deformations such as sinusoidal, Gaussian, and PLC band deformation, finding the reference image coordinate becomes a non-linear problem. For example, for PLC band deformation (Eq. 13), identifying reference coordinate x from the deformed image position x' is equivalent to solving non-linear equation

$$\Phi(x) = x + \frac{1}{2} ac\sqrt{\pi}\left[erf\left(\frac{x_0}{c_{plc}}\right) + erf\left(\frac{x - x_0}{c_{plc}}\right)\right] - x' = 0 \tag{14}$$

Solving non-linear equation is divided into two steps: initial value estimation and non-linear iteration.

The principle of initial value estimation is shown in Fig. 5; the aim is to estimate the reference position E' of point E in the deformed image. First, the deformed coordinates of reference integer positions are evaluated based on the given displacement field; for example, reference points c and d deform to c' and d' in the deformed image, respectively. Then, the interval that E is in is searched; as shown in Fig. 5, E is between c' and d'; since the deformation is continuous, E', the reference image position of E, should lie in interval (c, d). Finally, assuming the deformation in interval (c, d) is linear, the coordinate of E' can be estimated as

$$x(E') = x(c) + [x(d) - x(c)]\frac{x'(E) - x'(c')}{x'(d') - x'(c')}. \tag{15}$$

After estimating the initial value, Newton-Raphson method [26] is employed to solve the non-linear equation. For non-linear equation $\Phi(x) = 0$, the Newton-Raphson formula is

$$x_{k+1} = x_k - \frac{\Phi(x_k)}{\Phi'(x_k)}, \tag{16}$$

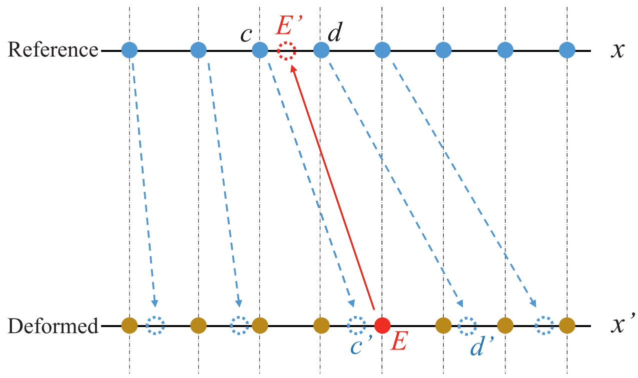

Fig. 5. Principle of initial value estimation.

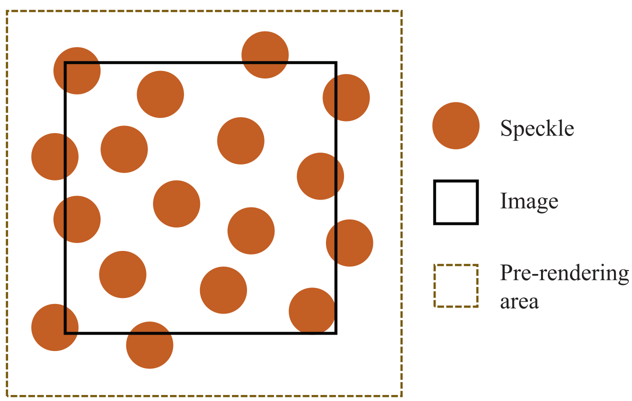

Fig. 6. Principle of speckle image pre-rendering technique.

where x_{k+1} and x_k denote the estimates at the $(k+1)$th and the kth iteration, respectively.

For example, for PLC band deformation (Eq. 13),

$$x_{k+1} = x_k - \frac{x_k + \frac{1}{2} ac\sqrt{\pi}\left[erf\left(\frac{x_0}{c_{\mathrm{plc}}}\right) + erf\left(\frac{x_k - x_0}{c_{\mathrm{plc}}}\right)\right] - x'}{1 + a\exp\left[-\frac{(x_k - x_0)^2}{c_{\mathrm{plc}}^2}\right]}. \tag{17}$$

The iteration is terminated when the iteration increment is less than 1×10^{-4}, and then the intensity is identified using the mathematical expression of Gaussian speckle pattern.

3.3. Pattern pre-rendering

The reference coordinate of deformed image's integer position is generally decimal; if the intensity is identified using interpolation methods, errors will be introduced inevitably. To avoid this error, *Glare* employs Gaussian speckle pattern, but the drawback of this choice is huge amount of computation. Since the software needs to display the deformed image real-time, the GUI will be stuck if each frame is rendered accurately. To improve GUI responsiveness, pattern pre-rendering technique is presented.

The principle of pre-rendering technique is illustrated in Fig. 6. Since the area that speckle exists is generally larger than the sampling area of reference image, we find an area that contains all speckles and generate a pre-rendering image: if a point is inside the pre-rendering image, its grayscale is estimated using linear interpolation; if the point is out of the pre-rendering area, the grayscale is set as background color. Thus, the calculation of Gaussian function is avoided, and the respondence speed of the graphical interface is significantly improved. Nevertheless, it is worth noting that pre-rending technique is used for graphical display exclusively; in the image export module, in order to ensure image quality, this technique is not used. In general, the process of rendering deformed image for non-linear deformation is depicted in Fig. 7.

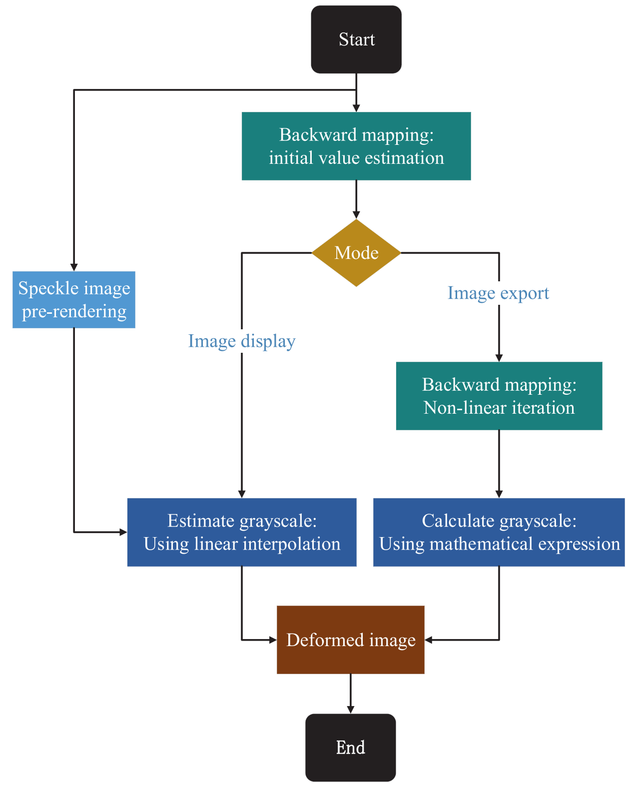

Fig. 7. Flowchart of deformed image rendering for non-linear deformation.

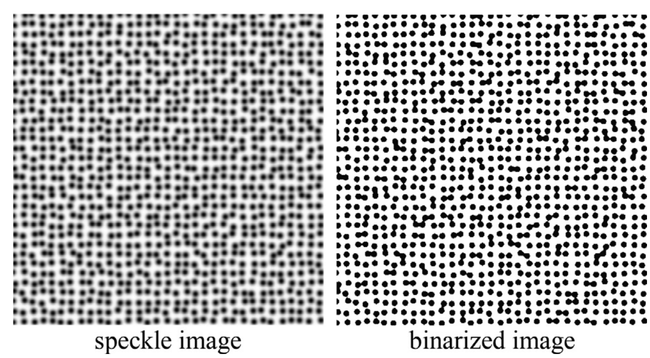

Fig. 8. Speckle image and corresponding binarized image.

4. Pattern quality assessment

Speckle image quality is of critical importance for digital image correlation method. Four important and widely used pattern quality assessment parameters are speckle coverage, speckle size, systematic error, and random error.

4.1. Speckle coverage

Speckle coverage characterizes the ratio of speckle area to the total area. A speckle coverage of 50% is generally considered appropriate.

After importing image, *Glare* will identify the binarization threshold using Ostu's method [27], binarize the speckle image (Fig. 8), calculate the ratio of black pixels, and show the speckle coverage automatically. For example, for the speckle image shown in Fig. 8, the binarization threshold by Ostu's algorithm is 148 and the coverage is 30.72%. The

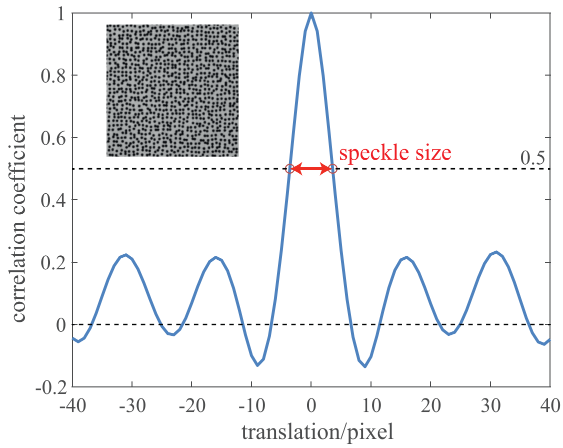

Fig. 9. Auto-correlation function and speckle size.

threshold can also be selected manually, and the software will automatically calculate the corresponding speckle coverage.

4.2. Speckle size

Speckle size characterizes speckle's length scale. Excessively small speckle size will cause image under-sampling, leading to large systematic error; excessively large speckle size will cause low image contrast, leading to large random error. Generally, it is believed that the optimal speckle size falls in the interval of 3–5 pixels.

Speckle size can be characterized by full width at half maxima of auto-correlation function [28], as shown in Fig. 9. The auto-correlation function is calculated as follows: first, select a subset centered at the image center and with given subset size; second, translate the original subset $[t, s]$ pixels to obtain a translated subset; third, calculate the correlation coefficient between the original subset and the translated subset, where the correlation coefficient is defined as

$$A_{ff}[t,s] = \frac{\sum_{m,n=-M}^{M} (f[m,n]-\bar{f})(f[m+t,n+s]-\bar{f})}{\sqrt{\sum_{m,n=-M}^{M} (f[m,n]-\bar{f})^2}\sqrt{\sum_{m,n=-M}^{M} (f[m+t,n+s]-\bar{f})^2}}, \tag{18}$$

here, $\bar{f}$ denotes the average of subset grayscales, t and s are translations along the x and y axes respectively, $2M+1$ is the subset size; finally, calculate the correlation coefficients for different translations and thus the auto-correlation function is obtained. *Glare* will identify the position where correlation coefficients equals 0.5 using linear interpolation and hence identify the speckle size. If the speckle sizes along the x and y axes are D_x and D_y respectively, the speckle size given by the software is

$$D = \sqrt{D_x D_y}. \tag{19}$$

4.3. Systematic error

Imperfect interpolation will cause a periodic systematic error [29], termed *interpolation bias*. Interpolation bias is mainly caused by high-frequency components of image. A small speckle size will cause significant high-frequency components, and thus large interpolation bias. High-order B-spline interpolation can effectively reduce the interpolation bias, but it has a drawback of increasing the computation burden; image filtering can also lower the interpolation bias, but it will reduce the spatial resolution.

The mathematical expression for interpolation bias of forward strategy is [30]

$$u_e \approx C_{ib} \sin 2\pi u_0, \tag{20}$$

where u_e denotes the interpolation bias, u_0 denotes the subpixel translation, C_{ib} denotes the amplitude of the interpolation bias, satisfying

$$C_{ib} = \frac{1}{2\pi}\frac{\int_{\frac{-1}{2}}^{\frac{1}{2}}\int_{\frac{-1}{2}}^{\frac{1}{2}} E_{ib}(v_x,v_y)\left|\hat{f}(v_x,v_y)\right|^2 \mathrm{d}v_x \mathrm{d}v_y}{\int_{\frac{-1}{2}}^{\frac{1}{2}}\int_{\frac{-1}{2}}^{\frac{1}{2}} v_x^2 \hat{\varphi}(v_x,v_y)\left|\hat{f}(v_x,v_y)\right|^2 \mathrm{d}v_x \mathrm{d}v_y}, \tag{21}$$

where $\hat{\varphi}(v_x, v_y)$ denotes the transfer function of the interpolation method, $|\hat{f}(v_x, v_y)|^2$ denotes the image power spectrum, $E_{ib}(v_x, v_y)$ denotes interpolation bias kernel, whose specific form is

$$\begin{aligned} E_{ib}(v_x,v_y) &= (v_x-1)\hat{\varphi}(v_x-1,v_y) - (v_x+1)\hat{\varphi}(v_x+1,v_y) \\ &\quad + \hat{\varphi}(v_x,v_y)\hat{\varphi}(v_x+1,v_y) + \hat{\varphi}(v_x,v_y)\hat{\varphi}(v_x-1,v_y). \end{aligned} \tag{22}$$

Glare will calculate the image power spectrum using discrete Fourier transform, estimate the interpolation bias amplitude using above equation, and show the results in the software interface. If the interpolation bias exceeds required accuracy, the interpolation bias should be reduced. From a view of algorithm, it can be realized by high-order B-spline interpolation or image pre-filtering technique; from a view of experiment, it can be achieved by increasing speckle physical size, decreasing object length, or improving camera resolution.

4.4. Random error

Image noise will cause fluctuations in measured displacements, termed random error. Large image noise will cause large random error; additionally, the random error is associated with image contrast.

The random error of zero- and first-order shape function is [31,32]

$$\mathrm{Std}(u) = \frac{\sqrt{2}\sigma}{\sqrt{\sum_{m=-M}^{M}\sum_{n=-M}^{M} f_x^2[m+m_0, n+n_0]}}, \tag{23}$$

where u denotes the measured displacement, σ denotes the standard deviation of image noise, $[m_0, n_0]$ denote the coordinates of subset center, $2M+1$ is the subset size, and $f_x[m,n]$ denotes the image gradient along the x axis at position $[m,n]$.

Interestingly, random error of second-order shape function *roughly* doubles that of zero- and first-order shape functions [33]. This phenomenon is caused by uniformity and isotropy of speckle pattern, and it is proved that the random error of second-order shape function is [34]

$$\mathrm{Std}(u) = \frac{\sqrt{7}\sigma}{\sqrt{\sum_{m=-M}^{M}\sum_{n=-M}^{M} f_x^2[m+m_0, n+n_0]}}. \tag{24}$$

Thus, the ratio of random errors corresponding second- and first-order shape functions equals to $\sqrt{7/2}$ (≈ 1.87), which is rather close to the empirical value 2.

Glare will evaluate the image gradient using Sobel operator, calculate sum of squared image gradients, estimate the random error using Eq. (23) or Eq. (24), and show the results in the software. If the random error is greater than the required precision, the random error must be reduced. There are two strategies to reduce the random error: (1) decreasing the image noise, such as using high-signal-to-noise-ratio camera or using multi-frame averaging technique (the shortcoming of multi-frame average denoising is that it cannot be used for dynamic deformation measurement); (2) increasing the sum of squared image gradient, such as using high contrast speckle pattern or increasing the subset size (the drawback of increasing subset size is that it leads to an decrease in spatial resolution). In real experiments, the selection of calculation parameters should consider systematic error, random error, and spatial resolution at the same time.

The most time-consuming part in estimating random error is evaluating

$$\sum_{m=-M}^{M}\sum_{n=-M}^{M} f_x^2[m+m_0, n+n_0]. \tag{25}$$

Conventional approach traverses the image and calculates the sum in Eq. (25) for each $[m_0, n_0]$; however, its computational efficiency is rather low due to the existence of redundant computation. To tackle this issue, *Glare* exploits *integral image* (or *summed area table*) method [35]. For given speckle image, an integral image

$$s(i,j) = \sum_{k=0}^{i} \sum_{l=0}^{j} f_x^2[k,l] \tag{26}$$

will be computed using a recursive algorithm

$$s[i,j] = s[i-1,j] + s[i,j-1] - s[i-1,j-1] + f_x^2[i,j]. \tag{27}$$

Thus, the sum in Eq. (25) can be calculated as

$$\begin{aligned}\sum_{m=-M}^{M} \sum_{m=-M}^{M} f_x^2[m+m_0, n+n_0] = \\ s[m_0+M, n_0+M] - s[m_0+M, n_0-M-1] \\ - s[m_0-M-1, n_0+M] + s[m_0-M-1, n_0-M-1].\end{aligned} \tag{28}$$

Through the use of integral image method, the computational efficiency of sum of squared image gradients is significantly improved.

5. Speckle pattern recommendation

With the emerging of water-transfer printing [36], thermal-mechanical toner transfer [37], and UV printing [38], fabricating high quality speckle pattern onto the specimen surface becomes feasible and has potential applications in single-camera 3D-DIC system [39] and high temperature deformation measurement [40]. Nowadays, DIC users can design high quality speckle pattern and then transfer the pattern to specimen surface, thereby improving the metrological performance.

Glare has a function of recommending speckle pattern for given field of view (FOV): the generated pattern is composed of randomly positioned disks with generation parameters of density $\rho = 60\%$, variation $\delta = 40\%$, and radius $r_i = 2$ pixels. Here, it is necessary to distinguish speckle image radius r_i and speckle physical radius r_o: the former, r_i, with a unit of pixel, represents the speckle size in the final image, while the later, r_o, with a unit of meter, characterizes the physical size of speckle on the specimen surface. Since the speckle image radius r_i is strongly correlated with experiment setup, parameters including object distance l_o, image distance l_i, and camera pixel size s needs to be given in the software. According to the pin-hole camera model,

$$r_i = \frac{l_i r_o}{l_o s}. \tag{29}$$

As r_i is set as 2 pixels, the speckle physical radius is

$$r_o = 2 l_o s / l_i. \tag{30}$$

Hence, a digital speckle pattern with radius of r_o, density of 60%, variation of 40%, and given field of view is generated, and this pattern can be exported as a vector image in format of pdf.

Additionally, the software has a function of suggesting camera resolution and object distance. (1) Suggesting camera resolution. In such a case, parameters including FOV, object distance, image distance, and pixel size are set; the software will automatically calculate the image size corresponding to the FOV. To image the whole FOV, the camera resolution must exceed the image width and image height given by the software. (2) Suggesting object distance. As the FOV is generally determined by the specimen to be measured, it can be regarded as preset; moreover, experimental apparatus are generally limited, and therefore camera parameters and lens focal length can be considered unchangeable; that is, only the object length can be adjusted. In such a case, user can firstly set the FOV, image distance (roughly equals to the lens focal length), pixel size, and then adjust the image height or image width; the software will calculate and show the corresponding object distance, facilitating the adjustment of experimental device.

6. Conclusion

This work reports a free and open-source software: *Glare* (homepage https://gitee.com/yongsu1989/glare). The software has functions including generating speckle patterns, rendering deformed images, assessing pattern quality, and presenting pattern recommendations. This software can generate ellipse, polygon, and Gaussian speckle patterns; can render deformed images with underlying deformation fields of translation, stretch/compression, rotation, sinusoidal deformation, Gaussian deformation, and PLC band deformation; can calculate key pattern quality assessment parameters such as speckle coverage, speckle size, systematic error, and random error; can produce optimized speckle pattern in form of vector image. In general, the software can be used not only for scientific research and engineering applications in digital image correlation community but also for education of experimental mechanic, and therefore has broad prospects.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yong Su: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Visualization, Funding acquisition. **Qingchuan Zhang:** Writing – review & editing, Project administration, Funding acquisition.

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